Topic 3: Singular Value Decomposition (SVD)

Question 1:

Given the matrix

A = [[3, 1],
[1, 3]]

Write a Python program using `np.linalg.svd()` to:

1. Compute U, S, and VT.
2. Construct the diagonal matrix Σ.
3. Verify the SVD by reconstructing A using A_reconstructed = U @ Sigma @ VT

Question 2:

Given the matrix

A = [[3, 1, 1],
[-1, 3, 1]]

Write a Python program to:

1. Compute the SVD of A.
2. Reconstruct A using UΣVT.
3. Calculate the reconstruction error using np.linalg.norm(A - A_reconstructed)

Question 3:

Given the matrix

A = [[4, 1, 2],
[2, 3, 1],
[1, 2, 5]]

Write a Python program to:

1. Compute the SVD of A using `np.linalg.svd()`.
2. Compute A^T A and find its eigenvalues using `np.linalg.eig()`.
3. Verify that the singular values obtained from SVD are the square roots of the eigenvalues of A^T A.
4. Print and compare the two results.